# EI331 Signals and Systems

## Lecture 31

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# Contents

# Unilateral Laplace Transform

Recall the (bilateral) Laplace transform of a CT signal $x$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

The unilateral Laplace transform of $x$ is

$$\mathcal{X}(s) = \int_{0_-}^{\infty} x(t) e^{-st} dt$$

also denoted

$$\mathcal{X} = \mathcal{UL}\{x\}, \quad x(t) \xleftrightarrow{\mathcal{UL}} \mathcal{X}(s)$$

ROC of $\mathcal{X}$ is **always** a **right** half-plane

NB. Information about $x(t)$ at $t < 0$ is lost in $\mathcal{X}$.

NB. $\mathcal{L}\{x\} = \mathcal{UL}\{x\}$ iff $x(t)$ is causal, i.e. $x(t) = 0$ for $t < 0$

# Examples

The calculation of unilateral Laplace transforms is almost the same as for bilateral Laplace transforms.

Example. $x_1(t) = e^{-at}u(t)$,

$$\mathcal{X}_1(s) = X_1(s) = \frac{1}{s+a}, \quad \text{Re}\, s > -\text{Re}\, a$$

Example. $x_2(t) = e^{-a(t+1)}u(t+1) = x_1(t+1)$,

$$\text{bilateral} \quad X_2(s) = e^s X_1(s) = \frac{e^s}{s+a}, \quad \text{Re}\, s > -\text{Re}\, a$$

$$\text{unilateral} \quad \mathcal{X}_2(s) = \int_0^\infty e^{-a(t+1)} e^{-st}\, dt = \frac{e^{-a}}{s+a}, \quad \text{Re}\, s > -\text{Re}\, a$$

NB. $X_2 \neq \mathcal{X}_2$, since $x_2$ is noncausal.

NB. $\mathcal{X}_2 \neq e^s \mathcal{X}_1$, time-shift property **fails** for unilateral transform.

# Examples

The calculation of unilateral Laplace transforms is almost the same as for bilateral Laplace transforms.

Example. $x_3(t) = e^{-a|t|}$ with $\operatorname{Re} a > 0$

$$\text{bilateral} \quad X_3(t) = \frac{-2a}{s^2 - a^2}, \quad -\operatorname{Re} a < \operatorname{Re} s < \operatorname{Re} a$$

$$\text{unilateral} \quad \mathcal{X}_3(s) = \frac{1}{s+a}, \quad \operatorname{Re} s > -\operatorname{Re} a$$

NB. $X_3 \neq X_1$, since $x_3 \neq x_1$

NB. $\mathcal{X}_3 = \mathcal{X}_1$, since $x_3$ and $x_1$ differ only for $t < 0$

Example. $x(t) = \delta(t) + 2\delta'(t) + e^t u(t)$

$$\mathcal{X}(s) = X(s) = 1 + 2s + \frac{1}{s-1}, \quad \operatorname{Re} s > 1$$

NB. The lower limit of integration is $0_-$ so $\delta$ and $\delta'$ contribute to $\mathcal{X}$

# Examples

The calculation of inverse unilateral Laplace transforms is the same as for bilateral Laplace transforms, but we can **only recover $x(t)$ for $t \geq 0$!**

Example. Given unilateral Laplace transform

$$\mathcal{X}(s) = \frac{1}{(s+1)(s+2)} = \frac{1}{s+1} - \frac{1}{s+2}$$

the **only** possibility for ROC is $\operatorname{Re} s > -1$. Thus

$$x(t) = e^{-t} - e^{-2t}, \quad t \geq 0$$

$\mathcal{X}$ provides **no** information about $x(t)$ for $t < 0$

Example. $\mathcal{X}(s) = \frac{s^2-3}{s+2} = -2 + s + \frac{1}{s+2}$. The ROC must be $\operatorname{Re} s > -2$, and

$$x(t) = -2\delta(t) + \delta'(t) + e^{-2t}, \quad t \geq 0$$

# Properties of Unilateral Laplace Transform

Many properties are the same as for bilateral Laplace transform

| Property | Signal | ULT |
|---|---|---|
| – | $x(t)$ | $\mathcal{X}(s)$ |
| – | $y(t)$ | $\mathcal{Y}(s)$ |
| Linearity | $ax(t) + by(t)$ | $a\mathcal{X}(s) + b\mathcal{Y}(s)$ |
| Shifting in $s$-domain | $e^{s_0 t}x(t)$ | $\mathcal{X}(s - s_0)$ |
| Time scaling | $x(at), a > 0$ | $\frac{1}{a}\mathcal{X}(\frac{s}{a})$ |
| Conjugation | $x^*(t)$ | $\mathcal{X}^*(s^*)$ |
| Differentiation in $s$ domain | $-tx(t)$ | $\frac{d}{ds}\mathcal{X}(s)$ |

# Convolution Property of Unilateral Laplace Transform

If $x(t) = y(t) = 0$ for $t < 0$, then

$$(x * y)(t) \xleftrightarrow{\mathcal{UL}} \mathcal{X}(s)\mathcal{Y}(s), \quad \text{ROAC} \supset \text{ROAC}_{\mathcal{X}} \cap \text{ROAC}_{\mathcal{Y}}$$

Proof. Note $(x * y)(t) = 0$ for $t < 0$.

$$\mathcal{UL}\{x * y\} = \mathcal{L}\{x * y\} = \mathcal{L}\{x\}\mathcal{L}\{y\} = \mathcal{UL}\{x\}\mathcal{UL}\{y\}$$

**Caution.** The assumption $x(t) = y(t) = 0$ for $t < 0$ is important!

Example. $x(t) = \delta(t) - \delta(t-1)$, $y(t) = \delta(t+1)$,

$$\mathcal{X}(s) = 1 - e^{-s}, \quad \mathcal{Y}(s) = 0,$$

Note $(x * y)(t) = \delta(t+1) - \delta(t)$, and

$$\mathcal{UL}\{x * y\} = -1 \neq \mathcal{X}(s)\mathcal{Y}(s)$$

# Integration in Time Domain

If

$$x(t) \overset{\mathcal{UL}}{\longleftrightarrow} \mathcal{X}(s), \quad \text{with ROAC} = R$$

then

$$\int_{0_-}^{t} x(\tau)d\tau \overset{\mathcal{UL}}{\longleftrightarrow} \frac{1}{s}\mathcal{X}(s), \quad \text{with ROAC} \supset R \cap \{\mathrm{Re}\, s > 0\}$$

Proof. Apply the convolution property to $\tilde{x} * u$, where

$$\tilde{x}(t) = \begin{cases} x(t), & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Example. $x(t) = \delta(t)$, $\mathcal{X}(s) = 1$.

$$\int_{0_-}^{t} x(\tau)d\tau = u(t) \overset{\mathcal{UL}}{\longleftrightarrow} \frac{1}{s}, \quad \mathrm{Re}\, s > 0$$

# Differentiation in Time Domain

If

$$x(t) \xleftrightarrow{\mathcal{UL}} \mathcal{X}(s), \quad \text{with ROC} = R$$

and $\lim_{t\to+\infty} x(t)e^{-st} = 0$ for $s \in R_0$, then

$$\frac{d}{dt}x(t) \xleftrightarrow{\mathcal{UL}} s\mathcal{X}(s) - x(0_-), \quad \text{with ROC} \supset R \cap R_0$$

NB. $R_0 \supset$ ROAC of $X(s)$ (also true for bilateral Laplace transform)

Proof. Use integration by parts

$$\int_{0_-}^{\infty} x'(t)e^{-st}dt = x(t)e^{-st}\Big|_{t=0_-}^{\infty} + s\int_{0_-}^{\infty} x(t)e^{-st}dt = -x(0_-) + s\mathcal{X}(s)$$

NB. Had we used the definition $\mathcal{X}(s) = \int_{0_+}^{\infty} x(t)e^{-st}dt$, we would have $\frac{d}{dt}x(t) \xleftrightarrow{\mathcal{UL}} s\mathcal{X}(s) - x(0_+)$

# Differentiation in Time Domain

Example. $x(t) = e^{-t}$, $\mathcal{X}(s) = \frac{1}{s+1}$ with ROAC $\mathrm{Re}\, s > -1$. By the differentiation property,

$$x'(t) \xleftrightarrow{\mathcal{UL}} \frac{s}{s+1} - 1 = -\frac{1}{s+1}, \quad \text{ROC} \supset \{\mathrm{Re}\, s > -1\}$$

By direct calculation,

$$x'(t) = -e^{-t} \xleftrightarrow{\mathcal{UL}} = -\frac{1}{s+1}, \quad \text{ROC} = \text{ROAC} = \{\mathrm{Re}\, s > -1\}$$

Example. $x(t) = e^{-t}u(t)$, $\mathcal{X}(s) = \frac{1}{s+1}$ with ROAC $\mathrm{Re}\, s > -1$. By the differentiation property,

$$x'(t) \xleftrightarrow{\mathcal{UL}} \frac{s}{s+1}, \quad \text{ROC} \supset \{\mathrm{Re}\, s > -1\}$$

By direct calculation, $x'(t) = \delta(t) - e^{-t}u(t)$, and

$$\mathcal{UL}\{x'\} = -\frac{1}{s+1} + 1 = \frac{s}{s+1}, \quad \text{ROC} = \text{ROAC} = \{\mathrm{Re}\, s > -1\}$$

# Differentiation in Time Domain

Under appropriate conditions, e.g. $x^{(k)}(t) = O(e^{\gamma t})$, we can extend the differentiation property to higher derivatives,

$$x^{(n)}(t) \xleftrightarrow{\mathcal{UL}} s^n \mathcal{X}(s) - \sum_{k=0}^{n-1} s^{n-1-k} x^{(k)}(0_-)$$

Since solutions to constant coefficient ODEs is of the form

$$x(t) = \sum_{k=1}^{n} p_i(t) e^{a_i t}$$

where $p_i$ are polynomials, the above condition is satisfied.

Unilateral Laplace transform is useful for solving ODEs with nonzero initial condtions.

# Initial Value Theorem

**Theorem.** If the unilateral Laplace transform $\mathcal{X}(s)$ of $x(t)$ is a proper rational function, then

$$x(0_+) = \lim_{s\to\infty} s\mathcal{X}(s)$$

**Proof.** $\mathcal{X}(s)$ has partial fraction expansion,

$$\mathcal{X}(s) = \sum_{i=1}^{r} \sum_{k_i=1}^{N_i} \frac{A_{i,k_i}}{(s+a_i)^{k_i}}$$

$$x(t) = \sum_{i=1}^{r} \sum_{k_i=1}^{N_i} \frac{A_{i,k_i} t^{k_i-1}}{(k_i-1)!} e^{-a_i t}, \quad t \geq 0$$

$$x(0_+) = \sum_{i=1}^{r} A_{i,1} = \lim_{s\to\infty} \sum_{i=1}^{r} \sum_{k_i=1}^{N_i} \frac{A_{i,k_i} s}{(s+a_i)^{k_i}} = \lim_{s\to\infty} s\mathcal{X}(s)$$

**Example.** $x(t) = 2e^{-t} + e^{-2t}$, $\mathcal{X}(s) = \frac{3s+5}{s^2+3s+2}$, $x(0_+) = \lim_{s\to\infty} s\mathcal{X}(s) = 3$.

# Initial Value Theorem

If $\mathcal{X}(s)$ is rational but not proper, then

$$\mathcal{X}(s) = \sum_{k=0}^{n} a_k s^k + \mathcal{X}_1(s)$$

where $\mathcal{X}_1(s)$ is a proper rational function.

Taking inverse transform,

$$x(t) = \sum_{k=0}^{n} a_k \delta^{(k)}(t) + x_1(t)$$

so

$$x(0_+) = x_1(0_+) = \lim_{s\to\infty} s\mathcal{X}_1(s)$$

Example. $x(t) = \delta(t) + 2e^{-t}$, $\mathcal{X}(s) = 1 + \frac{2}{s+1} = \frac{s+3}{s+1}$,

$$x(0_+) = 2 = \lim_{s\to\infty} \frac{2s}{s+1} \neq \lim_{s\to\infty} s\mathcal{X}(s)$$

# Example

Suppose a CT LTI system has the following properties.

1. The system is causal
2. The system function $H(s)$ is a proper rational function and has only two poles, at $s = -2$ and $s = -4$.
3. For input $x(t) = 1$, the output is $y(t) = 0$
4. The impulse response satisfies $h(0_+) = 4$

Determine the system function $H(s)$.

Solution. By 1 and 2

$$H(s) = \frac{as + b}{(s+2)(s+4)}, \quad \text{Re}\, s > -2$$

By 3, $H(0) = 0$, so $b = 0$.

By 4, $\lim_{s \to \infty} sH(s) = 4$, so $a = 4$, and

$$H(s) = \frac{4s}{(s+2)(s+4)}, \quad \text{Re}\, s > -2$$

# Final Value Theorem

**Theorem.** Suppose $x(t)$ does not contain $\delta$ or its derivatives for $t \geq 0$. If $\mathcal{X}(s)$ converges for real $s > 0$ and the $x(t)$ has a finite limit as $t \to +\infty$, then

$$\lim_{t\to+\infty} x(t) = \lim_{s\to 0_+} sX(s)$$

**Proof.** Since $A \triangleq \lim_{t\to+\infty} x(t)$ exists, $x(t)$ is bounded on $(0, +\infty)$, i.e. $|x(t)| \leq M$. For any $\epsilon > 0$, $\exists T$ s.t. $|x(t) - A| < \epsilon$ for $t \geq T$.

For $s > 0$,

$$s\mathcal{X}(s) - A = s\int_0^\infty [x(t) - A]e^{-st}dt$$

$$|s\mathcal{X}(s) - A| \leq s\left(\int_0^T + \int_T^\infty\right)|x(t) - A|e^{-st}dt = I_1 + I_2$$

where $I_1 \leq sT(M + A)$ and $I_2 \leq \epsilon s\int_T^\infty e^{-st}dt \leq \epsilon s\int_0^\infty e^{-st}dt = \epsilon$.

$$\lim_{s\to 0_+}|s\mathcal{X}(s) - A| \leq \epsilon \implies \lim_{s\to 0_+}|s\mathcal{X}(s) - A| = 0 \implies \lim_{s\to 0_+} s\mathcal{X}(s) = A$$

# Final Value Theorem

Example. $x(t) = 2 + 3e^{-t} + e^{-2t}$,

$$\mathcal{X}(s) = \frac{2}{s} + \frac{3}{s+1} + \frac{1}{s+2}, \quad \text{Re}\, s > 0$$

Check

$$\lim_{s \to 0_+} s\mathcal{X}(s) = 2 = \lim_{t \to +\infty} x(t)$$

The Final Values Theorem can be extended to allow finitely many $\delta$ and its derivatives in $x(t)$ on the positive real axis. We can rewrite $x(t)$ as

$$x(t) = x_1(t) + \sum_{i=1}^{n} a_i \delta^{(k_i)}(t - t_i) \xleftrightarrow{\mathcal{UL}} \mathcal{X}(s) = \mathcal{X}_1(s) + \sum_{i=1}^{n} a_i s^{k_i} e^{-st_i}$$

$$\lim_{t \to +\infty} x(t) = \lim_{t \to +\infty} x_1(t) = \lim_{s \to 0_+} s\mathcal{X}_1(s) = \lim_{s \to 0_+} s\mathcal{X}(s)$$

Example. $x(t) = \delta(t) + 2$, $\mathcal{X}(s) = \frac{1}{s} + 1$, $x(+\infty) = 2 = \lim_{s \to 0_+} s\mathcal{X}(s)$.

# Linear Constant-coefficient ODE

Consider ODE

$$\sum_{k=0}^{N} a_k \frac{d^k y}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x}{dt^k}$$

with initial condition $y^{(k)}(0_-)$, $k = 0, 1, \ldots, N-1$, and causal input, i.e. $x(t) = 0$ for $t < 0$.

Take unilateral Laplace transform of both sides

$$\sum_{k=0}^{N} a_k \left[ s^k \mathcal{Y}(s) - \sum_{\ell=0}^{k-1} s^{k-1-\ell} y^{(\ell)}(0_-) \right] = \sum_{k=0}^{M} b_k s^k \mathcal{X}(s)$$

so

$$\mathcal{Y}(s) = \underbrace{\frac{\sum_{k=0}^{M} b_k s^k}{\sum_{k=0}^{N} a_k s^k} \mathcal{X}(s)}_{\text{zero-state response}} + \underbrace{\frac{\sum_{k=0}^{N} a_k \sum_{\ell=0}^{k-1} s^{k-1-\ell} y^{(\ell)}(0_-)}{\sum_{k=0}^{N} a_k s^k}}_{\text{zero-input response}}$$

# Example

Consider ODE

$$y''(t) + 3y'(t) + 2y(t) = 2x'(t) + 6x(t)$$

with initial condition $y(0_-) = c_0, y'(0_-) = c_1$ and $x(t) = e^{-t}u(t)$.

Take unilateral Laplace transform of both sides

$$[s^2\mathcal{Y}(s) - sc_0 - c_1] + 3[s\mathcal{Y}(s) - c_0] + 2\mathcal{Y}(s) = 2s\mathcal{X}(s) + 6\mathcal{X}(s) = \frac{2s+6}{s+1}$$

so

$$\begin{aligned} \mathcal{Y}(s) &= \frac{2s+6}{(s^2+3s+2)(s+1)} + \frac{c_0 s + (3c_0 + c_1)}{s^2+3s+2} \\ &= \underbrace{\frac{2s+6}{(s+1)^2(s+2)}}_{\text{zero-state response}} + \underbrace{\frac{c_0 s + (3c_0 + c_1)}{(s+1)(s+2)}}_{\text{zero-input response}} \end{aligned}$$

# Example (cont'd)

## Zero-state response

$$\mathcal{Y}_{zs}(s) = \frac{2s+6}{(s+1)^2(s+2)}, \quad \text{Re } s > -1$$

For $t \geq 0$,

$$\begin{aligned} y_{zs}(t) &= \text{Res}[\mathcal{Y}_{zi} e^{st}, -2] + \text{Res}[\mathcal{Y}_{zi} e^{st}, -1] \\ &= \left.\frac{(2s+6)e^{st}}{(s+1)^2}\right|_{s=-2} + \left[\frac{d}{ds}\frac{(2s+6)e^{st}}{s+2}\right]_{s=-1} = 2e^{-2t} + (4t-2)e^{-t} \end{aligned}$$

## Zero-input response

$$\mathcal{Y}_{zi}(s) = \frac{c_0 s + (3c_0 + c_1)}{(s+1)(s+2)}, \quad \text{Re } s > -1$$

$$\begin{aligned} y_{zi}(t) &= \text{Res}[\mathcal{Y}_{zs} e^{st}, -2] + \text{Res}[\mathcal{Y}_{zs} e^{st}, -1] \\ &= (2c_0 + c_1)e^{-t} - (c_0 + c_1)e^{-2t}, \quad t \geq 0 \end{aligned}$$

# Contents

- Complex numbers and complex functions
  - arithmetic and representations
  - argument is multivalued, principal value $(-\pi, \pi]$
  - simply and multiply connected domains
  - continuity of functions
- Analytic functions
  - analytic at $z_0$ iff differentiable on some open disk $B(z_0, r)$
  - $f(z) = u(x, y) + jv(x, y)$ is analytic iff $u, v$ are continuously differentiable and satisfy Cauchy-Riemann equation

$$u_x = v_y, \quad u_y = -v_x$$

  - rational function $R(z) = \frac{N(z)}{D(z)}$ is analytic on $\mathbb{C}$ except for zeros of $D(z)$ (aka poles of $R(z)$)
  - elementary analytic functions

- Complex Integration
  - Cauchy's Theorem
$$\int_C f(z)dz = 0$$
  - Cauchy's Integral Formula
$$f^{(k)}(z) = \frac{n!}{j2\pi}\int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}d\zeta$$
- Series
  - power series and disk of convergence
  - Laurent series and annulus of convergence (cf. $z$-transform)
- Residue
  - $$\mathrm{Res}[f, z_0] = \begin{cases} \frac{1}{j2\pi}\int_C f(z)dz = c_{-1}, & z_0 \in \mathbb{C} \\ -\frac{1}{j2\pi}\int_C f(z)dz = -c_{-1}, & z_0 = \infty \end{cases}$$
  - Residue Theorem

- Signal Representations
  - time domain: linear combination of $\delta$, convolution
  - frequency domain: linear combination of sinusoids (CT/DT Fourier series/transforms), spectrum
  - $s$-domain: bilateral/unilateral Laplace transforms
  - $z$-domain: bilateral/unilateral $z$-transforms
- Sampling
  - Nyquist rate
  - spectra of sampled signals
- Systems
  - properties of systems (causality, stability, linearity, time-invariance...)
  - representations of LTI systems (differential/difference equations, block diagrams, impulse response, frequency response, system function)
  - eigenfunction property of LTI systems
  - filtering

- Transforms (Fourier, Laplace, $Z$)
    - forward and inverse transforms
    - properties
    - analysis of LTI systems
        - given system, find response to input
        - given input output pairs, identify system
- ODE
    - classical method in time domain
    - transform method
        - initial rest/LTI: Fourier, bilateral Laplace
        - nonzero initial condition: unilateral Laplace
- Difference equations
    - classical method in time domain
    - transform method
        - initial rest/LTI: Fourier, bilateral $z$-transform
        - nonzero initial condition: unilateral $z$-transform

Good Luck with Finals!